

CoRoT-1b

R-2477

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210205 · created 2026-08-23T23:52:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459250.7208 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.041
Transit depth (Rp/Rs)^2	1.65 +/- 1.05 [%]
Orbital Inclination (inc)	84.48 +/- 2.83
Airmass coefficient 1 (a1)	1.027 +/- 0.027
Airmass coefficient 2 (a2)	-0.018 +/- 0.054
Scatter in the residuals of the lightcurve fit is	2.65 %
Best Comparison Star	#3 - [384, 220]
Optimal Method	PSF photometry
Transit Duration (day)	0.1 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.129 ± 0.041 vs 0.1388 (deviation 0.21σ)
Duration fit vs published	0.1 d vs 0.1046 d (deviation 0.29σ)
Mid-time O–C	11.1 min
Depth SNR	2.8
Residual scatter	2.65 %

Light curve

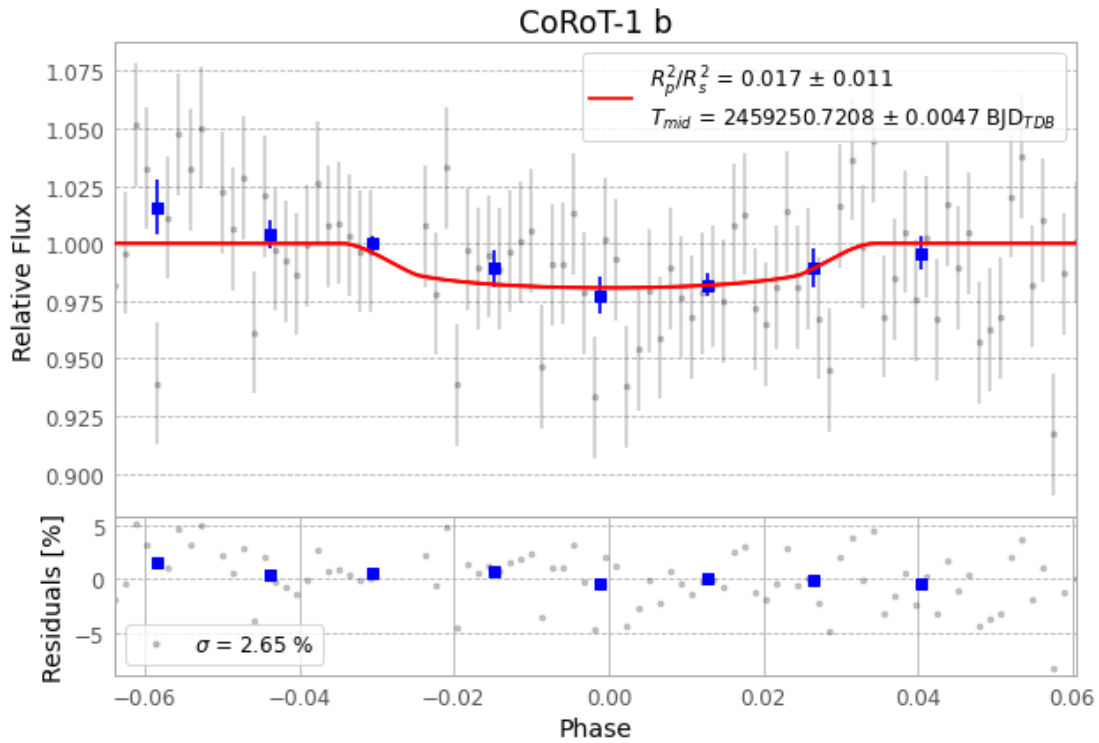


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-02-04.png (SHA-256 7d7e15ad2d88800c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7e97af585ff6ae7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

91 FITS frames (60 MB), CoRoT-1210205025112.FITS → CoRoT-1210205072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210205.log (b4ba7b740cd3...), FinalLightCurve_CoRoT-1 b_2021-02-04.png (7d7e15ad2d88...), FinalLightCurve_CoRoT-1 b_2021-02-04.csv (732b4897502c...)

Compute resources

Wall time 138.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:52Z.